

第十四课预习报告

book.c

- 作用: 展示如何用结构体存储并输出书籍信息。
- 功能: 定义 `struct book` , 读入书名、作者和价格, 最后格式化输出两种显示方式。
- 演示效果:

```
1  /* book.c -- one-book inventory */
2  #include <stdio.h>
3  #include <string.h>
4  char * s_gets(char * st, int n);
5  #define MAXTITL  41      /* maximum length of title + 1      */
6  #define MAXAUTL  31      /* maximum length of author's name + 1 */
7
8  struct book {           /* structure template: tag is book    */
9      char title[MAXTITL];
10     char author[MAXAUTL];
11     float value;
12 };                       /* end of structure template    */
13
14 int main(void)
15 {
16     struct book library; /* declare library as a book variable */
17
18     printf("Please enter the book title.\n");
19     s_gets(library.title, MAXTITL); /* access to the title portion */
20     printf("Now enter the author.\n");
21     s_gets(library.author, MAXAUTL);
22     printf("Now enter the value.\n");
23     scanf("%f", &library.value);
24     printf("%s by %s: $%.2f\n", library.title,
25           library.author, library.value);
26     printf("%s: \"%s\" ($%.2f)\n", library.author,
27           library.title, library.value);
28     printf("Done.\n");
29
30     return 0;
31 }
```

```

32
33 char * s_gets(char * st, int n)
34 {
35     char * ret_val;
36     char * find;
37
38     ret_val = fgets(st, n, stdin);
39     if (ret_val)
40     {
41         find = strchr(st, '\n');    // look for newline
42         if (find)                   // if the address is not NULL,
43             *find = '\0';           // place a null character there
44         else
45             while (getchar() != '\n')
46                 continue;           // dispose of rest of line
47     }
48     return ret_val;
49 }

```

```

1  Please enter the book title.
2  Now enter the author.
3  Now enter the value.
4  The C Programming Language by K&R: $59.99
5  K&R: "The C Programming Language" ($59.99)
6  Done.

```

booksave.c

- 作用: 用结构体数组存储多本书籍信息并保存至文件。
- 功能: 从文件读取已有书籍, 允许用户新增输入, 并将新书信息写入 `book.dat` 文件中。
- 演示效果:

```

1  /* booksave.c -- saves structure contents in a file */
2  #include <stdio.h>
3  #include <stdlib.h>
4  #include <string.h>
5  #define MAXTITL  40
6  #define MAXAUTL  40
7  #define MAXBKS   10           /* maximum number of books */
8  char * s_gets(char * st, int n);
9  struct book {                 /* set up book template */

```

```

10     char title[MAXTITL];
11     char author[MAXAUTL];
12     float value;
13 };
14
15 int main(void)
16 {
17     struct book library[MAXBKS]; /* array of structures */
18     int count = 0;
19     int index, filecount;
20     FILE * pbooks;
21     int size = sizeof (struct book);
22
23     if ((pbooks = fopen("book.dat", "a+b")) == NULL)
24     {
25         fputs("Can't open book.dat file\n", stderr);
26         exit(1);
27     }
28
29     rewind(pbooks);          /* go to start of file */
30     while (count < MAXBKS && fread(&library[count], size,
31                                     1, pbooks) == 1)
32     {
33         if (count == 0)
34             puts("Current contents of book.dat:");
35         printf("%s by %s: %.2f\n", library[count].title,
36               library[count].author, library[count].value);
37         count++;
38     }
39     filecount = count;
40     if (count == MAXBKS)
41     {
42         fputs("The book.dat file is full.", stderr);
43         exit(2);
44     }
45
46     puts("Please add new book titles.");
47     puts("Press [enter] at the start of a line to stop.");
48     while (count < MAXBKS && s_gets(library[count].title, MAXTITL) != NULL
49           && library[count].title[0] != '\0')
50     {
51         puts("Now enter the author.");
52         s_gets(library[count].author, MAXAUTL);
53         puts("Now enter the value.");

```

```

54     scanf("%f", &library[count++].value);
55     while (getchar() != '\n')
56         continue;          /* clear input line */
57     if (count < MAXBKS)
58         puts("Enter the next title.");
59 }
60
61 if (count > 0)
62 {
63     puts("Here is the list of your books:");
64     for (index = 0; index < count; index++)
65         printf("%s by %s: $%.2f\n", library[index].title,
66             library[index].author, library[index].value);
67     fwrite(&library[filecount], size, count - filecount,
68         pbooks);
69 }
70 else
71     puts("No books? Too bad.\n");
72
73 puts("Bye.\n");
74 fclose(pbooks);
75
76 return 0;
77 }
78
79 char * s_gets(char * st, int n)
80 {
81     char * ret_val;
82     char * find;
83
84     ret_val = fgets(st, n, stdin);
85     if (ret_val)
86     {
87         find = strchr(st, '\n');    // look for newline
88         if (find)                   // if the address is not NULL,
89             *find = '\0';           // place a null character there
90     else
91         while (getchar() != '\n')
92             continue;              // dispose of rest of line
93     }
94     return ret_val;
95 }

```

```
1 Current contents of book.dat:
2 Please add new book titles.
3 Press [enter] at the start of a line to stop.
4 Now enter the author.
5 Now enter the value.
6 Here is the list of your books:
7 New Book by Author X: $99.90
8 Bye.
```

complit.c

- 作用: 演示复合字面量 (compound literal) 的用法。
- 功能: 根据输入的分数选择不同的书籍信息, 并用复合字面量直接初始化结构体。
- 演示效果:

```
1  /* complit.c -- compound literals */
2  #include <stdio.h>
3  #define MAXTITL  41
4  #define MAXAUTL  31
5
6  struct book {          // structure template: tag is book
7      char title[MAXTITL];
8      char author[MAXAUTL];
9      float value;
10 };
11
12 int main(void)
13 {
14     struct book readfirst;
15     int score;
16
17     printf("Enter test score: ");
18     scanf("%d",&score);
19
20     if(score ≥ 84)
21         readfirst = (struct book) {"Crime and Punishment",
22                                     "Fyodor Dostoyevsky",
23                                     11.25};
24     else
25         readfirst = (struct book) {"Mr. Bouncy's Nice Hat",
26                                     "Fred Winsome",
```

```

27         5.99};
28     printf("Your assigned reading:\n");
29     printf("%s by %s: $%.2f\n", readfirst.title,
30           readfirst.author, readfirst.value);
31
32     return 0;
33 }

```

```

1   Enter test score: Your assigned reading:
2   Crime and Punishment by Fyodor Dostoyevsky: $11.25

```

enum.c

- 作用: 演示枚举类型和字符串比较的使用。
- 功能: 用户输入颜色名, 根据匹配结果输出对应颜色描述, 否则提示无法识别。
- 演示效果:

```

1   /* enum.c -- uses enumerated values */
2   #include <stdio.h>
3   #include <string.h>    // for strcmp(), strchr()
4   #include <stdbool.h>   // C99 feature
5   char * s_gets(char * st, int n);
6
7   enum spectrum {red, orange, yellow, green, blue, violet};
8   const char * colors[] = {"red", "orange", "yellow",
9     "green", "blue", "violet"};
10  #define LEN 30
11
12  int main(void)
13  {
14      char choice[LEN];
15      enum spectrum color;
16      bool color_is_found = false;
17
18      puts("Enter a color (empty line to quit):");
19      while (s_gets(choice, LEN) != NULL && choice[0] != '\0')
20      {
21          for (color = red; color ≤ violet; color++)
22          {
23              if (strcmp(choice, colors[color]) == 0)
24              {

```

```

25         color_is_found = true;
26         break;
27     }
28 }
29 if (color_is_found)
30     switch(color)
31     {
32         case red    : puts("Roses are red.");
33             break;
34         case orange : puts("Poppies are orange.");
35             break;
36         case yellow : puts("Sunflowers are yellow.");
37             break;
38         case green  : puts("Grass is green.");
39             break;
40         case blue   : puts("Bluebells are blue.");
41             break;
42         case violet : puts("Violets are violet.");
43             break;
44     }
45     else
46         printf("I don't know about the color %s.\n", choice);
47     color_is_found = false;
48     puts("Next color, please (empty line to quit):");
49 }
50 puts("Goodbye!");
51
52 return 0;
53 }
54
55 char * s_gets(char * st, int n)
56 {
57     char * ret_val;
58     char * find;
59
60     ret_val = fgets(st, n, stdin);
61     if (ret_val)
62     {
63         find = strchr(st, '\n');    // look for newline
64         if (find)                   // if the address is not NULL,
65             *find = '\0';           // place a null character there
66     }
67     else
68         while (getchar() != '\n')
69             continue;               // dispose of rest of line

```

```
69     }
70     return ret_val;
71 }
```

```
1  Enter a color (empty line to quit):
2  Roses are red.
3  Next color, please (empty line to quit):
4  Bluebells are blue.
5  Next color, please (empty line to quit):
6  I don't know about the color black.
7  Next color, please (empty line to quit):
8  Goodbye!
```

flexmemb.c

- 作用: 演示 C99 引入的“柔性数组成员” (flexible array member) 特性。
- 功能: 展示动态分配结构体与可变长度数组的结合使用。
- 演示效果:

```
1  // flexmemb.c -- flexible array member (C99 feature)
2  #include <stdio.h>
3  #include <stdlib.h>
4
5  struct flex
6  {
7      size_t count;
8      double average;
9      double scores[]; // flexible array member
10 };
11
12 void showFlex(const struct flex * p);
13
14 int main(void)
15 {
16     struct flex * pf1, *pf2;
17     int n = 5;
18     int i;
19     int tot = 0;
20
21     // allocate space for structure plus array
22     pf1 = malloc(sizeof(struct flex) + n * sizeof(double));
```



```

23     pf1->count = n;
24     for (i = 0; i < n; i++)
25     {
26         pf1->scores[i] = 20.0 - i;
27         tot += pf1->scores[i];
28     }
29     pf1->average = tot / n;
30     showFlex(pf1);
31
32     n = 9;
33     tot = 0;
34     pf2 = malloc(sizeof(struct flex) + n * sizeof(double));
35     pf2->count = n;
36     for (i = 0; i < n; i++)
37     {
38         pf2->scores[i] = 20.0 - i/2.0;
39         tot += pf2->scores[i];
40     }
41     pf2->average = tot / n;
42     showFlex(pf2);
43     free(pf1);
44     free(pf2);
45
46     return 0;
47 }
48
49 void showFlex(const struct flex * p)
50 {
51     int i;
52     printf("Scores : ");
53     for (i = 0; i < p->count; i++)
54         printf("%g ", p->scores[i]);
55     printf("\nAverage: %g\n", p->average);
56 }

```

```

1 Scores : 20 19 18 17 16
2 Average: 18
3 Scores : 20 19.5 19 18.5 18 17.5 17 16.5 16
4 Average: 17.8889

```

- 作用: 展示结构体嵌套与初始化的典型用法。
- 功能: 定义两个结构体—— `names` 与 `guy` , 其中一个嵌套在另一个中, 实现层次化数据管理并格式化输出。
- 演示效果:

```
1 // friend.c -- example of a nested structure
2 #include <stdio.h>
3 #define LEN 20
4 const char * msgs[5] =
5 {
6     "    Thank you for the wonderful evening, ",
7     "You certainly prove that a ",
8     "is a special kind of guy. We must get together",
9     "over a delicious ",
10    " and have a few laughs"
11 };
12
13 struct names {                // first structure
14     char first[LEN];
15     char last[LEN];
16 };
17
18 struct guy {                  // second structure
19     struct names handle;      // nested structure
20     char favfood[LEN];
21     char job[LEN];
22     float income;
23 };
24
25 int main(void)
26 {
27     struct guy fellow = {     // initialize a variable
28         { "Ewen", "Villard" },
29         "grilled salmon",
30         "personality coach",
31         68112.00
32     };
33
34     printf("Dear %s, \n\n", fellow.handle.first);
35     printf("%s%s.\n", msgs[0], fellow.handle.first);
36     printf("%s%s\n", msgs[1], fellow.job);
37     printf("%s\n", msgs[2]);
38     printf("%s%s%s", msgs[3], fellow.favfood, msgs[4]);
```

```

39     if (fellow.income > 150000.0)
40         puts("!!");
41     else if (fellow.income > 75000.0)
42         puts("!");
43     else
44         puts(".");
45     printf("\n%40s%s\n", " ", "See you soon,");
46     printf("%40s%s\n", " ", "Shalala");
47
48     return 0;
49 }

```

```

1   Dear Ewen,
2
3       Thank you for the wonderful evening, Ewen.
4   You certainly prove that a personality coach
5   is a special kind of guy. We must get together
6   over a delicious grilled salmon and have a few laughs.
7
8                               See you soon,
9   Shalala

```

friends.c

- 作用: 演示指向结构体的指针及其成员访问方式 (-> 与 *运算符的区别)。
- 功能: 展示如何通过结构体指针访问成员并计算地址偏移。
- 演示效果:

```

1   /* friends.c -- uses pointer to a structure */
2   #include <stdio.h>
3   #define LEN 20
4
5   struct names {
6       char first[LEN];
7       char last[LEN];
8   };
9
10  struct guy {
11      struct names handle;
12      char favfood[LEN];
13      char job[LEN];

```

```

14     float income;
15 };
16
17 int main(void)
18 {
19     struct guy fellow[2] = {
20         {{ "Ewen", "Villard"},
21            "grilled salmon",
22            "personality coach",
23            68112.00
24         },
25         {"Rodney", "Swillbelly"},
26            "tripe",
27            "tabloid editor",
28            432400.00
29     };
30 };
31 struct guy * him;    /* here is a pointer to a structure */
32
33 printf("address #1: %p #2: %p\n", &fellow[0], &fellow[1]);
34 him = &fellow[0];    /* tell the pointer where to point */
35 printf("pointer #1: %p #2: %p\n", him, him + 1);
36 printf("him->income is $%.2f: (*him).income is $%.2f\n",
37        him->income, (*him).income);
38 him++;                /* point to the next structure */
39 printf("him->favfood is %s: him->handle.last is %s\n",
40        him->favfood, him->handle.last);
41
42 return 0;
43 }

```

```

1  address #1: 000000000061FE10 #2: 000000000061FE60
2  pointer #1: 000000000061FE10 #2: 000000000061FE60
3  him->income is $68112.00: (*him).income is $68112.00
4  him->favfood is tripe: him->handle.last is Swillbelly

```

funct_ptr.c

- 作用: 演示函数指针的使用及动态指定函数行为。
- 功能: 用户输入字符串后, 通过菜单选择不同的函数 (大写、小写、大小写互换、原样输出) 来处理字符串。

- 演示效果:

```
1 // func_ptr.c -- uses function pointers
2 #include <stdio.h>
3 #include <string.h>
4 #include <ctype.h>
5 #define LEN 81
6 char * s_gets(char * st, int n);
7 char showmenu(void);
8 void eatline(void); // read through end of line
9 void show(void (* fp)(char *), char * str);
10 void ToUpper(char *); // convert string to uppercase
11 void ToLower(char *); // convert string to lowercase
12 void Transpose(char *); // transpose cases
13 void Dummy(char *); // leave string unaltered
14
15 int main(void)
16 {
17     char line[LEN];
18     char copy[LEN];
19     char choice;
20     void (*pfun)(char *); // points a function having a
21     // char * argument and no
22     // return value
23     puts("Enter a string (empty line to quit):");
24     while (s_gets(line, LEN) != NULL && line[0] != '\0')
25     {
26         while ((choice = showmenu()) != 'n')
27         {
28             switch (choice) // switch sets pointer
29             {
30                 case 'u' : pfun = ToUpper; break;
31                 case 'l' : pfun = ToLower; break;
32                 case 't' : pfun = Transpose; break;
33                 case 'o' : pfun = Dummy; break;
34             }
35             strcpy(copy, line); // make copy for show()
36             show(pfun, copy); // use selected function
37         }
38         puts("Enter a string (empty line to quit):");
39     }
40     puts("Bye!");
41
42     return 0;
```

```
43     }
44
45     // ... (略部分函数定义, 见源文件)
```

```
1   Enter a string (empty line to quit):
2   Enter menu choice:
3   u) uppercase          l) lowercase
4   t) transposed case o) original case
5   n) next string
6   TEST_STRING
7   test_string
8   tEST_sTRING
9   Test_String
10  Enter a string (empty line to quit):
11  Bye!
```

funds1.c

- 作用: 演示将结构体成员作为函数参数传递的用法。
- 功能: 定义 `struct funds` 存储银行和储蓄账户信息, 通过 `sum()` 函数计算总资金。
- 演示效果:

```
1   /* funds1.c -- passing structure members as arguments */
2   #include <stdio.h>
3   #define FUNDLLEN 50
4
5   struct funds {
6       char    bank[FUNDLLEN];
7       double  bankfund;
8       char    save[FUNDLLEN];
9       double  savefund;
10  };
11
12  double sum(double, double);
13
14  int main(void)
15  {
16      struct funds stan = {
17          "Garlic-Melon Bank",
18          4032.27,
19          "Lucky's Savings and Loan",
```

```

20         8543.94
21     };
22
23     printf("Stan has a total of $%.2f.\n",
24           sum(stan.bankfund, stan.savefund) );
25     return 0;
26 }
27
28 /* adds two double numbers */
29 double sum(double x, double y)
30 {
31     return(x + y);
32 }

```

```

1 | Stan has a total of $12576.21.

```

funds4.c

- 作用: 演示将结构体数组作为函数参数传递的用法。
- 功能: 展示如何处理包含多个结构体的数组并对其成员数据求和。
- 演示效果:

```

1  /* funds4.c -- passing an array of structures to a function */
2  #include <stdio.h>
3  #define FUNLEN 50
4  #define N 2
5
6  struct funds {
7      char    bank[FUNLEN];
8      double  bankfund;
9      char    save[FUNLEN];
10     double  savefund;
11 };
12
13 double sum(const struct funds money[], int n);
14
15 int main(void)
16 {
17     struct funds jones[N] = {
18         {
19             "Garlic-Melon Bank",

```

```

20         4032.27,
21         "Lucky's Savings and Loan",
22         8543.94
23
24     },
25     {
26         "Honest Jack's Bank",
27         3620.88,
28         "Party Time Savings",
29         3802.91
30     }
31 };
32
33 printf("The Joneses have a total of $%.2f.\n",
34        sum(jones,N));
35
36 return 0;
37 }
38
39 double sum(const struct funds money[], int n)
40 {
41     double total;
42     int i;
43
44     for (i = 0, total = 0; i < n; i++)
45         total += money[i].bankfund + money[i].savefund;
46
47     return(total);
48 }

```

1 | The Joneses have a total of \$20000.00.

manybook.c

- 作用: 演示如何使用结构体数组存储和管理多本书籍的信息。
- 功能: 支持逐本输入书名、作者与价格, 最后按录入顺序输出书籍清单。
- 演示效果:

```

1  /* manybook.c -- multiple book inventory */
2  #include <stdio.h>
3  #include <string.h>

```



```

4 char * s_gets(char * st, int n);
5 #define MAXTITL 40
6 #define MAXAUTL 40
7 #define MAXBKS 100 /* maximum number of books */
8
9 struct book { /* set up book template */
10     char title[MAXTITL];
11     char author[MAXAUTL];
12     float value;
13 };
14
15 int main(void)
16 {
17     struct book library[MAXBKS]; /* array of book structures */
18     int count = 0;
19     int index;
20
21     printf("Please enter the book title.\n");
22     printf("Press [enter] at the start of a line to stop.\n");
23     while (count < MAXBKS && s_gets(library[count].title, MAXTITL) != NULL
24           && library[count].title[0] != '\0')
25     {
26         printf("Now enter the author.\n");
27         s_gets(library[count].author, MAXAUTL);
28         printf("Now enter the value.\n");
29         scanf("%f", &library[count++].value);
30         while (getchar() != '\n')
31             continue; /* clear input line */
32         if (count < MAXBKS)
33             printf("Enter the next title.\n");
34     }
35
36     if (count > 0)
37     {
38         printf("Here is the list of your books:\n");
39         for (index = 0; index < count; index++)
40             printf("%s by %s: %.2f\n", library[index].title,
41                   library[index].author, library[index].value);
42     }
43     else
44         printf("No books? Too bad.\n");
45
46     return 0;
47 }

```

```

48
49 char * s_gets(char * st, int n)
50 {
51     char * ret_val;
52     char * find;
53
54     ret_val = fgets(st, n, stdin);
55     if (ret_val)
56     {
57         find = strchr(st, '\n');    // look for newline
58         if (find)                    // if the address is not NULL,
59             *find = '\0';           // place a null character there
60         else
61             while (getchar() != '\n')
62                 continue;          // dispose of rest of line
63     }
64     return ret_val;
65 }

```

```

1  Please enter the book title.
2  Press [enter] at the start of a line to stop.
3  Now enter the author.
4  Now enter the value.
5  Enter the next title.
6  Now enter the author.
7  Now enter the value.
8  Here is the list of your books:
9  The C Programming Language by K&R: $59.99
10 C Primer Plus by Stephen Prata: $75.50

```

names1.c

- 作用: 演示如何使用结构体指针访问和修改结构体成员。
- 功能: 通过指针传递结构体地址, 实现输入姓名、计算姓名字母总数并输出结果。
- 演示效果:

```

1  /* names1.c -- uses pointers to a structure */
2  #include <stdio.h>
3  #include <string.h>
4
5  #define NLEN 30

```

```

6  struct namect {
7      char fname[NLEN];
8      char lname[NLEN];
9      int letters;
10 };
11
12 void getinfo(struct namect *);
13 void makeinfo(struct namect *);
14 void showinfo(const struct namect *);
15 char * s_gets(char * st, int n);
16
17 int main(void)
18 {
19     struct namect person;
20
21     getinfo(&person);
22     makeinfo(&person);
23     showinfo(&person);
24     return 0;
25 }
26
27 void getinfo (struct namect * pst)
28 {
29     printf("Please enter your first name.\n");
30     s_gets(pst→fname, NLEN);
31     printf("Please enter your last name.\n");
32     s_gets(pst→lname, NLEN);
33 }
34
35 void makeinfo (struct namect * pst)
36 {
37     pst→letters = strlen(pst→fname) +
38     strlen(pst→lname);
39 }
40
41 void showinfo (const struct namect * pst)
42 {
43     printf("%s %s, your name contains %d letters.\n",
44         pst→fname, pst→lname, pst→letters);
45 }
46
47 char * s_gets(char * st, int n)
48 {
49     char * ret_val;

```

```

50     char * find;
51
52     ret_val = fgets(st, n, stdin);
53     if (ret_val)
54     {
55         find = strchr(st, '\n');    // look for newline
56         if (find)                  // if the address is not NULL,
57             *find = '\0';          // place a null character there
58         else
59             while (getchar() != '\n')
60                 continue;          // dispose of rest of line
61     }
62     return ret_val;
63 }

```

```

1   Please enter your first name.
2   Please enter your last name.
3   John Doe, your name contains 7 letters.

```

names2.c

- 作用: 演示结构体的按值传递与返回。
- 功能: 展示如何将结构体作为函数返回值和参数传递, 实现姓名信息的输入、处理和输出。
- 演示效果:

```

1   /* names2.c -- passes and returns structures */
2   #include <stdio.h>
3   #include <string.h>
4
5   #define NLEN 30
6   struct namect {
7       char fname[NLEN];
8       char lname[NLEN];
9       int letters;
10  };
11
12  struct namect getinfo(void);
13  struct namect makeinfo(struct namect);
14  void showinfo(struct namect);
15  char * s_gets(char * st, int n);
16

```

```

17 int main(void)
18 {
19     struct namect person;
20
21     person = getinfo();
22     person = makeinfo(person);
23     showinfo(person);
24
25     return 0;
26 }
27
28 struct namect getinfo(void)
29 {
30     struct namect temp;
31     printf("Please enter your first name.\n");
32     s_gets(temp.fname, NLEN);
33     printf("Please enter your last name.\n");
34     s_gets(temp.lname, NLEN);
35
36     return temp;
37 }
38
39 struct namect makeinfo(struct namect info)
40 {
41     info.letters = strlen(info.fname) + strlen(info.lname);
42
43     return info;
44 }
45
46 void showinfo(struct namect info)
47 {
48     printf("%s %s, your name contains %d letters.\n",
49           info.fname, info.lname, info.letters);
50 }
51
52 char * s_gets(char * st, int n)
53 {
54     char * ret_val;
55     char * find;
56
57     ret_val = fgets(st, n, stdin);
58     if (ret_val)
59     {
60         find = strchr(st, '\n');    // look for newline

```

```

61         if (find)                // if the address is not NULL,
62             *find = '\0';        // place a null character there
63     else
64         while (getchar() != '\n')
65             continue;            // dispose of rest of line
66     }
67     return ret_val;
68 }

```

```

1   Please enter your first name.
2   Please enter your last name.
3   Alice Smith, your name contains 10 letters.

```

names3.c

- 作用: 演示如何使用动态内存分配 (malloc/free) 来存储结构体成员字符串。
- 功能: 通过 `malloc()` 动态分配内存保存姓和名, 计算总字母数并输出, 然后释放内存。
- 演示效果:

```

1   // names3.c -- use pointers and malloc()
2   #include <stdio.h>
3   #include <string.h>    // for strcpy(), strlen()
4   #include <stdlib.h>    // for malloc(), free()
5   #define SLEN 81
6   struct namect {
7       char * fname;    // using pointers
8       char * lname;
9       int  letters;
10  };
11
12  void getinfo(struct namect *);    // allocates memory
13  void makeinfo(struct namect *);
14  void showinfo(const struct namect *);
15  void cleanup(struct namect *);    // free memory when done
16  char * s_gets(char * st, int n);
17
18  int main(void)
19  {
20      struct namect person;
21
22      getinfo(&person);

```

```

23     makeinfo(&person);
24     showinfo(&person);
25     cleanup(&person);
26
27     return 0;
28 }
29
30 void getinfo (struct namect * pst)
31 {
32     char temp[SLEN];
33     printf("Please enter your first name.\n");
34     s_gets(temp, SLEN);
35     // allocate memory to hold name
36     pst->fname = (char *) malloc(strlen(temp) + 1);
37     // copy name to allocated memory
38     strcpy(pst->fname, temp);
39     printf("Please enter your last name.\n");
40     s_gets(temp, SLEN);
41     pst->lname = (char *) malloc(strlen(temp) + 1);
42     strcpy(pst->lname, temp);
43 }
44
45 void makeinfo (struct namect * pst)
46 {
47     pst->letters = strlen(pst->fname) +
48     strlen(pst->lname);
49 }
50
51 void showinfo (const struct namect * pst)
52 {
53     printf("%s %s, your name contains %d letters.\n",
54           pst->fname, pst->lname, pst->letters);
55 }
56
57 void cleanup(struct namect * pst)
58 {
59     free(pst->fname);
60     free(pst->lname);
61 }
62
63 char * s_gets(char * st, int n)
64 {
65     char * ret_val;
66     char * find;

```

```

67
68     ret_val = fgets(st, n, stdin);
69     if (ret_val)
70     {
71         find = strchr(st, '\n');    // look for newline
72         if (find)                    // if the address is not NULL,
73             *find = '\0';           // place a null character there
74         else
75             while (getchar() != '\n')
76                 continue;           // dispose of rest of line
77     }
78     return ret_val;
79 }

```

```

1  Please enter your first name.
2  Please enter your last name.
3  Bob Brown, your name contains 8 letters.

```

funds3.c

- 作用: 演示将整个结构体作为函数参数传递的用法。
- 功能: 展示结构体按值传递的机制, 通过 `sum()` 函数接收结构体副本计算总资金。
- 演示效果:

```

1  /* funds3.c -- passing a structure */
2  #include <stdio.h>
3  #define FUNDLLEN 50
4
5  struct funds {
6      char    bank[FUNDLLEN];
7      double  bankfund;
8      char    save[FUNDLLEN];
9      double  savefund;
10 };
11
12 double sum(struct funds moolah); /* argument is a structure */
13
14 int main(void)
15 {
16     struct funds stan = {
17         "Garlic-Melon Bank",

```



```

18         4032.27,
19         "Lucky's Savings and Loan",
20         8543.94
21     };
22
23     printf("Stan has a total of $%.2f.\n", sum(stan));
24
25     return 0;
26 }
27
28 double sum(struct funds moolah)
29 {
30     return(moolah.bankfund + moolah.savefund);
31 }

```

```

1 | Stan has a total of $12576.21.

```

funds2.c

- 作用: 演示将结构体指针作为函数参数传递的用法。
- 功能: 定义 `struct funds` 存储银行和储蓄账户信息, 通过 `sum()` 函数接收结构体指针并计算总资金。
- 演示效果:

```

1  /* funds2.c -- passing a pointer to a structure */
2  #include <stdio.h>
3  #define FUNDLLEN 50
4
5  struct funds {
6      char    bank[FUNDLLEN];
7      double  bankfund;
8      char    save[FUNDLLEN];
9      double  savefund;
10 };
11
12 double sum(const struct funds *); /* argument is a pointer */
13
14 int main(void)
15 {
16     struct funds stan = {
17         "Garlic-Melon Bank",

```

```
18         4032.27,  
19         "Lucky's Savings and Loan",  
20         8543.94  
21     };  
22  
23     printf("Stan has a total of $%.2f.\n", sum(&stan));  
24  
25     return 0;  
26 }  
27  
28 double sum(const struct funds * money)  
29 {  
30     return(money->bankfund + money->savefund);  
31 }
```

```
1 | Stan has a total of $12576.21.
```

```
1 |
```